

Exercise 4 Mhluri Maswanganye

```
(6) SELECT account_id,  
          account_holder,  
          'Sandton' AS city  
FROM branch-sandton-accounts  
UNION  
SELECT account_id,  
          account_holder,  
          'Rosebank' AS city  
FROM branch-rosebank-accounts;
```

account_id	account_holder	city
A001	Nomvula Dlamini	Sandton
A002	David Mokoena	Sandton
A003	Lerato Sithole	Rosebank
A004	Lerato Sithole	Sandton
A005	Peter Nkosi	Rosebank
A006	Peter Nkosi	Sandton
A007	Zanele Khumalo	Rosebank
A008	Thabo Mthema	Rosebank

```
(7) SELECT product_code,  
          product_name,  
          'savings' AS product_type  
FROM saving-products  
UNION  
SELECT product_code,  
          product_name,  
          'current' AS product_type  
FROM current-products;
```

product_code	product_name	product_type
CR01	Standard Current	Current
CR02	Gold Current	Current
CR03	Business Current	Current
SV01	Basic Savings	Savings
SV02	Premium Savings	Savings
SV03	Youth Saving	Current
SV03	Youth Saving	Savings
SV04	Business Saving	Savings

```

(3) SELECT staff-id,
        staff-name,
        CONCAT(LOWER(staff-id), '@netbank.co.za') AS email
FROM retail-banking-staff
UNION
SELECT staff-id,
        staff-name,
        CONCAT(LOWER(staff-id), '@netbank.co.za') AS email
FROM corporate-banking-staff;

```

staff-id	staff-name	email
S001	Mpho Radobo	S001@netbank.co.za
S002	Brian Tshabalala	S002@netbank.co.za
S003	Aisha Patel	S003@netbank.co.za
S004	Kabolo Moabelo	S004@netbank.co.za
S005	Nandi Dube	S005@netbank.co.za
S006	Sipho Khumalo	S006@netbank.co.za

```

(29) SELECT city-code,
           city-name,
           'Mobile Branch' AS region
FROM mobile-branch-cities
UNION
SELECT city-code,
       city-name,
       'Digital Branch' AS region
FROM digital-branch-cities;

```

city-code	city-name	region
CO1	Johannesburg	Digital Branch
CO1	Johannesburg	Mobile Branch
CO2	Pretoria	Mobile Branch
CO3	Cape town	Digital Branch
CO3	Cape town	Mobile Branch
CO4	Durban	Mobile Branch
CO5	Polokwane	Digital Branch
CO6	Port Elizabeth	Digital Branch

```

(30) SELECT customer-id,
           customer-name,
           'Push Notification' AS segment
FROM push-notification-targets
UNION
SELECT customer-id,
       customer-name,
       'In-App Banner' AS segment
FROM inapp-banner-targets;

```


customer-id	customer-name	segment
C1001	Nomsa Zwane	Push Notification
C1002	Andile Buthelezi	Push Notification
C1003	Andile Buthelezi	In-App Banner
C1003	Fatima Mohamed	Push Notification
C1003	Fatima Mahomed	In-App Banner
C1004	Ryno van Zyl	Push Notification
C1005	Thandeka Cole	In-App Banner
C1006	Samuel Nkansi	In-App Banner

```

(6) SELECT txn_id AS transaction_id,
       account_id,
       amount,
       '2026-07-01' AS transaction_date
FROM atm01-transactions

SELECT txn_id AS transaction_id,
       account_id,
       amount,
       '2026-07-02' AS transaction_date
FROM atm02-transaction;

```

transaction_id	account-id	amount	transactional-date
T1001	A001	500.00	2026-07-01
T1002	A002	1200.00	2026-07-01
T1003	A001	300.00	2026-07-01
T1004	A003	750.00	2026-07-01
T1003	A001	300.00	2026-07-02
T1005	A004	900.00	2026-07-02
T1006	A002	450.00	2026-07-02
T1007	A005	150.00	2026-07-02

⑥ SELECT app_id As application_id,
 customer_id,
 loan_type,
 500000.00 As amount_requested
 FROM gauteng_loan_applications
 UNION ALL
 SELECT app_id As application_id,
 customer_id,
 loan_type,
 350000.00 As amount_requested
 FROM western_cape_loan_application;

application_id	customer_id	loan_type	amount_requested
LA001	C1001	Home Loan	
LA002	C1002	Vehicle Loan	
LA003	C1003	Personal	
LA004	C1004	Home Loan	
LA003	C1003	Personal	
LA005	C1005	Vehicle Loan	
LA006	C1006	Business	
LA007	C1007	Home Loan	

⑧ SELECT complaint_id,
 customer_id,
 'Email' As category,
 logged_date
 FROM email_complaints
 UNION ALL

SELECT complaint_id,
 customer_id,
 'Mobile App' As category,
 logged_date
 FROM app_complaints;

complaint_id	customer_id	category	logged_date
EC001	C2001	Email	2026-07-02
EC002	C2002	Email	2026-07-05
EC003	C2003	Email	2026-07-10
EC004	C2004	Email	2026-07-14
AC001	C2005	Mobile App	2026-07-23
AC002	C2001	Mobile App	2026-07-08
AC003	C2006	Mobile App	2026-07-16
AC004	C2007	Mobile App	2026-07-20

(Q9) SELECT payment_id,
 account_id,
 amount,
 payment_date
 FROM april_payments
 UNION ALL
 SELECT payment_id,
 account_id,
 amount,
 payment_date
 FROM may_payments

Payment_id	account_id	amount	payment_date
PAY001	A001	12500.00	2026-04-03
PAY002	A002	4800.00	2026-04-08
PAY003	A003	9200.00	2026-04-15
PAY004	A004	3300.00	2026-04-27
PAY005	A001	12500.00	2026-05-02
PAY006	A005	7600.00	2026-05-09
PAY007	A002	5100.00	2026-05-18
PAY008	A006	2800.00	2026-05-25

```

⑩ SELECT entry-id,
        account-id,
        entry-type,
        amount,
        entry-date
FROM debit-entries
UNION ALL
SELECT entry-id,
        account-id,
        entry-type,
        amount,
        entry-date
FROM credit-entries;

```

entry-id	account-id	entry-type	amount	entry-date
DR001	A001	Debit	1250.00	2026-06-02
DR002	A002	Debit	850.00	2026-06-05
DR003	A003	Debit	2300.00	2026-06-10
DR004	A004	Debit	1750.00	2026-06-15
CR001	A001	Credit	500.00	2026-06-03
CR002	A005	Credit	3200.00	2026-06-08
CR003	A002	Credit	1100.00	2026-06-12
CR004	A006	Credit	2750.00	2026-06-18

⑫ Bonus Question

- ① UNION, because it combines both the January and February watch list and removes duplicate records
- ② UNION ALL, because it keeps all records
- ③ The same number of columns & same order must be used to make the query work